

Title & Authors

Game Title: DIO

Authors: Hoi Yin Li & Emily Kuang

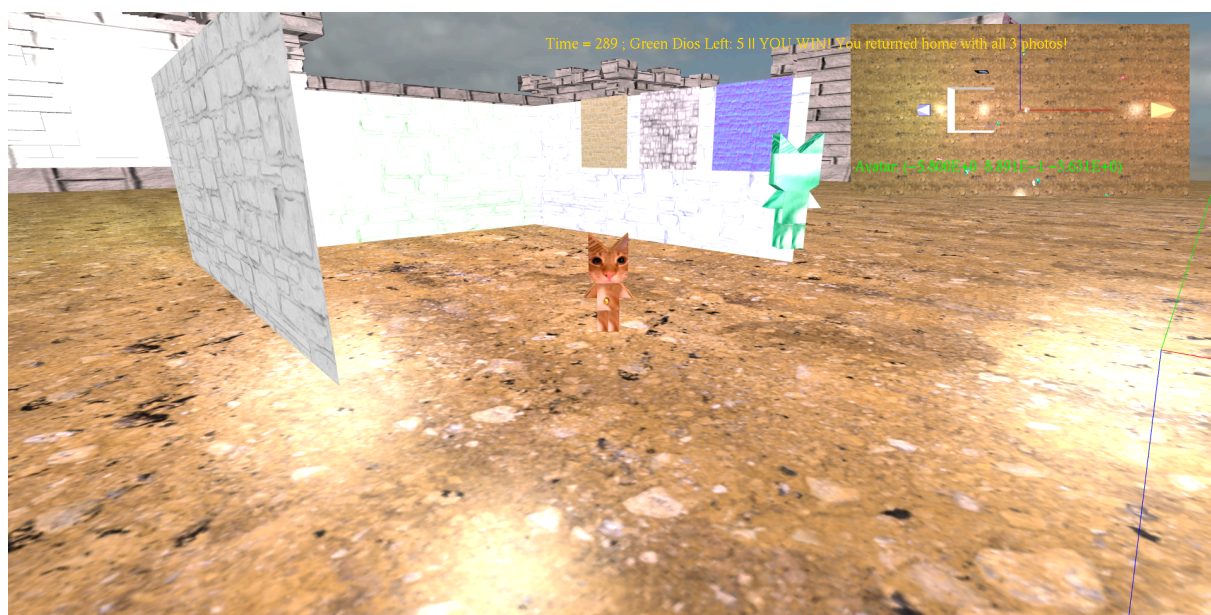
Section: CSC 165 – Section 02

Semester: Spring 2026

Gameplay Screenshot



Dio & Miku on a server together with green Dio NPC's in view.



Secret Easter Egg: Bringing home all 3 pyramid photos will still let you win.

How to Compile & Run Game

For Running Network Server: `run_server`

For Playing as Miku: `run_miku`

For Playing as Dio: `run_dio`

How to Play & Scoring System

The player explores a dynamic 3D landscape controlling either Dio or Miku, navigating a game world by five green Dio and 2 predatory red Dio which they randomly walk in the game. Because of the physics implementation, dio and miku will be walking like sliding and it will be more challenging. Either avatar is near the green dio and the green dio will stop moving. The player must kill all green Dio in order to win the game. Each player can go near the green Dio and press "SPACE" to hit the green dio. When there is 0 green dio left, you win the game. However, if the player collides with a red dio, it will trigger a game over.

Player Controls

KEY	ACTION
W	Move avatar forwards
A	Rotate avatar left
S	Move avatar backwards
D	Rotate avatar right
Z	Zoom main avatar camera inwards
X	Zoom main avatar camera outwards
I	Move overhead mini-map camera forwards
J	Move overhead mini-map camera left
K	Move overhead mini-map camera backwards
L	Move overhead mini-map camera right
N	Zoom overhead mini-map camera inwards
M	Zoom overhead mini-map camera outwards
SPACE	Hit/beat green cats
Q	Turn on/off the light

E	Toggle viewing of physics capsules
UP ARROW	Orbit main avatar camera upwards
DOWN ARROW	Orbit main avatar camera downwards
LEFT ARROW	Orbit main avatar camera left
RIGHT ARROW	Orbit main avatar camera right
P	Takes a photo of a pyramid when in range
R	Reset avatar manually
LEFT STICK Y_AXIS	Move avatar forwards/backwards
LEFT STICK X_AXIS	Rotate avatar left/right

Lighting

The game uses multiple dynamic lights. A red-tinted light above Pyramid 1, green-tinted light above Pyramid 2, a blue-tinted light above Pyramid 3, and a purple-tinted light in Dio's home. Playable avatars will emit a soft light attached too. Using the Q-key toggles all lights (excluding the avatar's) on and off, with the avatar's light state being the alternate of the other, in order for the game to still be visible.

Networking

- Create NPC: When a new client connects and requests data for an NPC, the server transmits an NPC packet (index 1 for red Dio, index 0 for green Dio). This allows the client to visually bind the correct texture for the color of the Dio.
- Check (isnr /isnear): the server will track the distance location between a player and NPC.
- Synchronized the deletion of NPC and global scoring.

Additions to TAGE Engine

[M] Modified, [+] Added/created, * Not mine (aka TAGE's)

[+] OverheadPanAction.java

[+] OverheadZoomAction.java

[M] KeyboardTurnLeftAction.java

[M] KeyboardTurnRightAction.java

[+] BackAction.java

[M] FwdAction.java

[M] TurnAction.java

[M] YawAction.java

[+] Pyramid.java
[M] DioHouse.java
[+] PulseController.java
[+] StretchController.java
[+] CameraOrbit3D.java
[M] Camera.java*
[M] Viewport.java*
[M] GameObject.java*
[M] JOALAudioManager.java*

Genre, Theme, Dimensionality, & Activities

Genre: Action / Rescue

Theme: Survival & exploration

Dimensionality: 3D world with physics-based movement

Activity: Combat-killing green Dio, avoiding red Dio, exploring & navigating world.

Location of Project Requirements

External Models: We have two model (Miku and Dio)

Networked Multi-player: We are using UDP for networking. run_dio for select dio as character; run_miku for select miku as character

Skybox and Terrain: Sky dome with day sky texture attached to avatar and heightmap terrain has grass texture in the corner of the map.

Light: press Q for turn on/ off the light

HUD: it will display how many dolphin left and win/lose game message

3D Sounds: the sound is from the dolphin_logo

Animation: Dio's walk(press WASD) and hit(press SPACE); Miku walk(press WS) and hit(press SPACE)

NPCs: there are 5 green dio and 2 red dio (They both walk randomly)

Physics: Player slides on terrain; collisions with red Dio trigger Game Over.

Nonfunctional Project Requirements

None

Special Requirements

None

Team Contribution(s)

Member	Contribution(s)
Hoi Yin Li	Game logic, networking, Miku animation/skeleton/OBJ, code merging
Emily Kuang	3D audio sounds, physics, terrain, skybox, Dio animation/skeleton/OBJ

Assets & Credits

Asset Name	Credits	Copyright Licenses
blue_brick.JPG	https://ambientcg.com/view?id=Bricks084	https://docs.ambientcg.com/license/
brick1.JPG	Dr. Scott Gordon	https://athena.ecs.csus.edu/~gordonvs/
day_sky.JPG	Day Sky HDRI 026 A ambientCG	https://docs.ambientcg.com/license/
Dolphin_HighPolyUV.JPG	Dr. Scott Gordon	https://athena.ecs.csus.edu/~gordonvs/
marble_brick.JPG	https://ambientcg.com/view?id=Bricks084	https://docs.ambientcg.com/license/
rocky_brick.JPG	https://ambientcg.com/view?id=Bricks084	https://docs.ambientcg.com/license/
rocky_ground.JPG	https://ambientcg.com/view?id=Ground104	https://docs.ambientcg.com/license/
sand_brick.JPG	https://ambientcg.com/view?id=Bricks084	https://docs.ambientcg.com/license/
warped_gray.JPG	https://ambientcg.com/view?id=Ground104	https://docs.ambientcg.com/license/
dolphin_logo.PNG	Emily Kuang	https://kleki.com/#licenses
miku.PNG	Hoi Yin Li	N/A
dio.PNG	Emily Kuang	https://kleki.com/#licenses
dio_green.PNG	Emily Kuang	https://kleki.com/#licenses
dio_red.PNG	Emily Kuang	https://kleki.com/#licenses
grass.JPG	https://ambientcg.com/a/Grass004	https://docs.ambientcg.com/license/
hills.JPG	https://ambientcg.com/a/Rock063	https://docs.ambientcg.com/license/
attackSfx.WAV	https://www.myinstants.com/en/instant/bonk-doge-81934/	https://www.myinstants.com/en/dcma_copyright_policy.html
bgMusic.WAV	https://pixabay.com/music/upbeat-game-minecraft-gaming-background-music-402451/ Music by Maksym Malko from Pixabay	https://pixabay.com/service/license-summary/
dolphinSfx.WAV	https://www.myinstants.com/en/instant/dolphin/	https://www.myinstants.com/en/dcma_copyright_policy.html
castle.obj	https://poly.pizza/m/opTOMcN3o9 Castle by Quaternius	https://creativecommons.org/publicdomain/zero/1.0/

Lab Machines Tested

ECS-SPACEINVADE, ECS-SONIC